ILT Research System - Architecture & Technical Infrastructure Specification (v5.5)

Component Architecture, Full Relational Data Pipeline, DTH-A116 240Hz Calibration, Restored Per-Block & Video Records, and Ethics Data Isolation (Consolidated Edition)

Document Parameter	Specification Value	Document Parameter	Specification Value
Document Identifier:	ILT-ARCH-SPEC-V5.5-EN	Publication Date:	October 22, 2026
Functional Baseline:	FRS v3.4 & DDD v1.5	Digitizer Sampling Rate:	240Hz (270Hz Peak, 0.1ms us timing)
Target Digitizer:	Wacom MovinkPad 11 (DTH-A116)	Geometric Calibration:	5.195 CSS px/mm, mm coordinates
Mirroring Latency:	<15ms via pointerrawupdate	Data Governance:	Identity Isolation, Privacy by Default

1. System Architectural Overview & Segmented Data Pipelines (v5.5)

Architecture v5.5 implements all technical corrections from the October 22, 2026 Consolidated Gap Report. It establishes strict segregation between real-time screen mirroring (<15ms), reliable telemetry persistence into TimescaleDB, overhead desk video relay via go2rtc, restored session_blocks and session_video_records tables, and child data isolation.

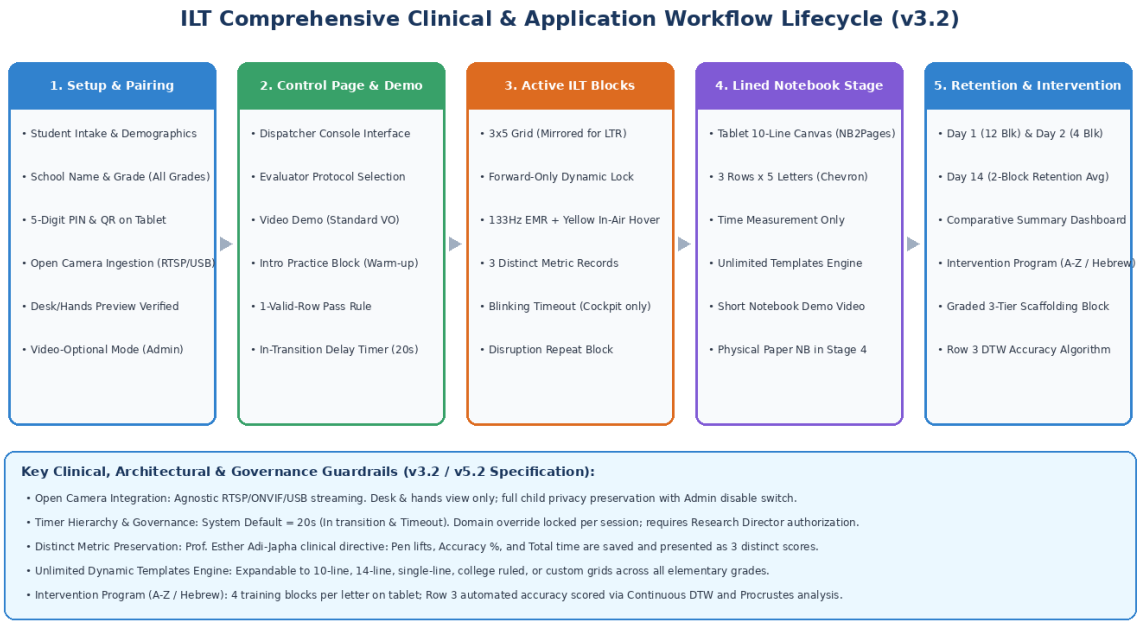


Figure 1: Comprehensive System Architecture, Network Topology & Data Pipeline

2. Production Database DDL Specification (PostgreSQL 16 + TimescaleDB v1.5)

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-- ILT RESEARCH APPLICATION PLATFORM - PRODUCTION DATABASE DDL (v1.5)
-- Engine: PostgreSQL 16 + TimescaleDB Extension (DTH-A116 240Hz Calibrated)
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CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS timescaledb CASCADE;

-- 1. System Users & RBAC Roles (Dev Accounts Standardized, O1)
CREATE TABLE IF NOT EXISTS users (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    email VARCHAR(255) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    full_name VARCHAR(100) NOT NULL,
    role VARCHAR(50) NOT NULL CHECK (role IN ('RESEARCHER', 'EVALUATOR', 'ANALYST', 'ADMIN')),
    director_auth_pin_hash VARCHAR(255),
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- 2. Sensitive Participant Identities (Strict Restricted Access, P2)
CREATE TABLE IF NOT EXISTS participant_identities (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    national_id_hash VARCHAR(64) UNIQUE NOT NULL,
    full_legal_name VARCHAR(100) NOT NULL,
    birth_date DATE NOT NULL,
    school_name VARCHAR(150) NOT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 3. De-Identified Research Participants Entity (P2, P3, P4)
CREATE TABLE IF NOT EXISTS participants (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    identity_ref_id UUID NULL REFERENCES participant_identities(id) ON DELETE SET NULL,
    study_id VARCHAR(50) UNIQUE NOT NULL, -- Anonymous study code (e.g. STUDY-2026-TLV-042)
    school_id_hash VARCHAR(64) NOT NULL,  -- Keyed HMAC-SHA256 (P3)
    age_months INT NOT NULL,               -- Age recorded at registration (P2)
    gender VARCHAR(20) NOT NULL CHECK (gender IN ('MALE', 'FEMALE', 'OTHER')),
    grade_level VARCHAR(20) NOT NULL CHECK (grade_level IN ('KINDERGARTEN', 'GRADE_1', 'GRADE_2',
'GRADE_3', 'GRADE_4', 'GRADE_5', 'GRADE_6')),
    socioeconomic_status VARCHAR(20) NOT NULL CHECK (socioeconomic_status IN ('LOW_SES', 'MID_SES',
'HIGH_SES')),
    dominant_hand VARCHAR(20) NOT NULL CHECK (dominant_hand IN ('RIGHT', 'LEFT', 'AMBIDEXTROUS')),
    consent_signed_date DATE NOT NULL,
    ethics_approval_ref VARCHAR(100) NOT NULL, -- Helsinki / Ministry of Education approval (P4)
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 4. Parental Consent Records (Privacy by Default, P1)
CREATE TABLE IF NOT EXISTS consent_records (
    consent_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    participant_id UUID NOT NULL REFERENCES participants(id) ON DELETE CASCADE,
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    consent_version VARCHAR(20) NOT NULL,
    allows_hand_video_recording BOOLEAN NOT NULL DEFAULT FALSE, -- Privacy by default! (P1)
    allows_anonymized_open_science BOOLEAN NOT NULL DEFAULT FALSE, -- Privacy by default! (P1)
    signed_at TIMESTAMPTZ DEFAULT NOW()
);

-- 5. Global System Configuration (20s Defaults, 240Hz, Calibration)
CREATE TABLE IF NOT EXISTS system_config (
    config_id INT PRIMARY KEY DEFAULT 1,
    in_transition_delay_sec INT NOT NULL DEFAULT 20,
    inactivity_timeout_sec INT NOT NULL DEFAULT 20,
    post_block_timeout_sec INT NOT NULL DEFAULT 30,
    notebook_page_timeout_sec INT NOT NULL DEFAULT 60,
    digitizer_sampling_rate_hz INT NOT NULL DEFAULT 240, -- D1: 240Hz design rate
    display_calibration_px_per_mm DECIMAL(6,3) NOT NULL DEFAULT 5.195, -- D3: 5.195 px/mm
    display_ppi DECIMAL(5,1) NOT NULL DEFAULT 230.9,
    video_recording_enabled BOOLEAN NOT NULL DEFAULT FALSE,
    require_research_director_approval BOOLEAN NOT NULL DEFAULT TRUE,
    score_weight_pen_lifts_pct DECIMAL(5,2) NOT NULL DEFAULT 35.00,
    score_weight_accuracy_pct DECIMAL(5,2) NOT NULL DEFAULT 40.00,
    score_weight_time_pct DECIMAL(5,2) NOT NULL DEFAULT 25.00,
    updated_at TIMESTAMPTZ DEFAULT NOW(),
    CONSTRAINT chk_score_weights_sum CHECK (
        score_weight_pen_lifts_pct + score_weight_accuracy_pct + score_weight_time_pct = 100.00
    )
);

-- 6. Domains (Districts)
CREATE TABLE IF NOT EXISTS domains (
    domain_id SERIAL PRIMARY KEY,
    domain_name VARCHAR(100) UNIQUE NOT NULL,
    override_in_transition_sec INT NULL,
    override_inactivity_timeout_sec INT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 7. Canonical Letter Models (Real ILT Chevron Shape, B7)
CREATE TABLE IF NOT EXISTS letter_canonical_models (
    model_id VARCHAR(50) PRIMARY KEY,
    letter_symbol VARCHAR(20) NOT NULL,
    script_type VARCHAR(30) NOT NULL CHECK (script_type IN ('ILT_INVENTED', 'HEBREW_STANDARD',
'ARABIC_STANDARD', 'ENGLISH_STANDARD')),
    target_dots_json JSONB NOT NULL,       -- 3 target dots (entry, apex, terminus)
    nominal_width_mm DECIMAL(4,1) NOT NULL DEFAULT 15.0,
    width_tolerance_mm DECIMAL(4,1) NOT NULL DEFAULT 2.0,
    canonical_polyline JSONB NOT NULL,      -- Array of [x_mm, y_mm] coordinates
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 8. Assessment Templates
CREATE TABLE IF NOT EXISTS dynamic_templates (
    template_id VARCHAR(50) PRIMARY KEY,
    template_name VARCHAR(150) NOT NULL,
    template_category VARCHAR(50) NOT NULL,
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    ruling_type VARCHAR(50) NOT NULL,
    canvas_schema JSONB NOT NULL,
    rules_schema JSONB NOT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 9. Diagnostic Sessions (Supporting Day 1/2/14 and Intervention Stages 1-4, R6)
CREATE TABLE IF NOT EXISTS diagnostic_sessions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_code VARCHAR(20) UNIQUE NOT NULL,
    participant_id UUID NOT NULL REFERENCES participants(id) ON DELETE RESTRICT,
    evaluator_id UUID NOT NULL REFERENCES users(id) ON DELETE RESTRICT,
    domain_id INT NOT NULL REFERENCES domains(domain_id),
    session_type VARCHAR(30) NOT NULL CHECK (session_type IN (
        'DAY_1_BASELINE', 'DAY_2_RETENTION', 'DAY_14_TRANSFER',
        'INTERVENTION_STAGE_1', 'INTERVENTION_STAGE_2', 'INTERVENTION_STAGE_3',
'INTERVENTION_STAGE_4'
    )),
    status VARCHAR(50) NOT NULL DEFAULT 'INITIALIZED' CHECK (status IN ('INITIALIZED', 'PAIRED',
'IN_PROGRESS', 'PAUSED', 'COMPLETED', 'ABORTED')),
    active_in_transition_sec INT NOT NULL DEFAULT 20,
    active_inactivity_sec INT NOT NULL DEFAULT 20,
    is_timer_overridden BOOLEAN NOT NULL DEFAULT FALSE,
    sync_status VARCHAR(50) NOT NULL DEFAULT 'LOCAL_BUFFER',
    started_at TIMESTAMPTZ DEFAULT NOW(),
    completed_at TIMESTAMPTZ NULL
);

-- 10. Per-Block Results Table (Restored, R1)
CREATE TABLE IF NOT EXISTS session_blocks (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_id UUID NOT NULL REFERENCES diagnostic_sessions(id) ON DELETE CASCADE,
    block_index SMALLINT NOT NULL,
    attempt_number SMALLINT NOT NULL DEFAULT 1,
    template_id VARCHAR(50) NOT NULL REFERENCES dynamic_templates(template_id),
    box15_completed BOOLEAN NOT NULL DEFAULT TRUE,
    skipped_boxes_count SMALLINT NOT NULL DEFAULT 0,
    raw_duration_ms INT NOT NULL DEFAULT 0,
    active_writing_time_ms INT NOT NULL DEFAULT 0,
    air_time_ms INT NOT NULL DEFAULT 0,
    total_pen_lifts_in_block INT NOT NULL DEFAULT 0,
    internal_pen_lifts_count INT NOT NULL DEFAULT 0,
    accuracy_percentage DECIMAL(5,2) NOT NULL DEFAULT 0.00,
    dots_hit_percentage DECIMAL(5,2) NOT NULL DEFAULT 0.00,
    width_compliance_percentage DECIMAL(5,2) NOT NULL DEFAULT 0.00,
    dtw_trajectory_score DECIMAL(5,2) NOT NULL DEFAULT 0.00,
    is_valid BOOLEAN NOT NULL DEFAULT TRUE,
    is_repeated BOOLEAN NOT NULL DEFAULT FALSE,
    sha256_batch_hash VARCHAR(64) NULL,
    created_at TIMESTAMPTZ DEFAULT NOW(),
    CONSTRAINT unq_session_block_attempt UNIQUE (session_id, block_index, attempt_number)
);

-- 11. Desk Video Records Table (Restored, R2)
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CREATE TABLE IF NOT EXISTS session_video_records (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_id UUID NOT NULL REFERENCES diagnostic_sessions(id) ON DELETE CASCADE,
    video_uri VARCHAR(512) NOT NULL,
    camera_angle VARCHAR(50) NOT NULL DEFAULT 'OVERHEAD_HANDS',
    sync_offset_ms INT NOT NULL DEFAULT 0,
    sha256_hash VARCHAR(64) NOT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 12. Pen Telemetry Hypertable (REAL/float4, Microseconds, Active Box, R4, R5, D2)
CREATE TABLE IF NOT EXISTS pen_telemetry_chunks (
    session_id UUID NOT NULL REFERENCES diagnostic_sessions(id) ON DELETE CASCADE,
    block_index SMALLINT NOT NULL,
    active_box_id SMALLINT NOT NULL DEFAULT 0, -- R5: active box id
    time_offset_us BIGINT NOT NULL,            -- D2: microseconds
    x_mm REAL NOT NULL,                        -- D3, R4: millimeters (float4)
    y_mm REAL NOT NULL,                        -- D3, R4: millimeters (float4)
    pressure REAL NOT NULL,                    -- D8: normalized pressure [0.0 - 1.0]
    tilt_x SMALLINT NOT NULL DEFAULT 0,
    tilt_y SMALLINT NOT NULL DEFAULT 0,
    stroke_flags SMALLINT NOT NULL DEFAULT 0,  -- R3: bitmask
    recorded_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

SELECT create_hypertable('pen_telemetry_chunks', 'recorded_at', if_not_exists => TRUE, migrate_data
=> TRUE);

CREATE INDEX IF NOT EXISTS idx_telemetry_session_block_time
    ON pen_telemetry_chunks (session_id, block_index, time_offset_us);

-- 13. De-Identified Analyst Research View (Zero PII, Full Kinematics & Demographics)
CREATE OR REPLACE VIEW v_analyst_cohort_slices AS
SELECT
    s.id AS session_id,
    p.study_id AS deidentified_study_code,
    p.school_id_hash,
    p.grade_level,
    p.socioeconomic_status,
    p.dominant_hand,
    p.age_months,
    p.gender,
    s.session_type,
    b.block_index,
    b.attempt_number,
    b.raw_duration_ms,
    b.active_writing_time_ms,
    b.air_time_ms,
    b.total_pen_lifts_in_block,
    b.internal_pen_lifts_count,
    b.accuracy_percentage,
    b.dots_hit_percentage,
    b.width_compliance_percentage,
    b.dtw_trajectory_score,
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    b.is_valid,
    s.started_at,
    s.completed_at
FROM diagnostic_sessions s
JOIN participants p ON s.participant_id = p.id
JOIN session_blocks b ON b.session_id = s.id
WHERE s.status = 'COMPLETED';
```

3. Hardware Calibration & 240Hz Sampling Pipeline (D1–D10)

- **240Hz Digitizer Sampling & pointerrawupdate Mirroring:** Directly binds to pointerrawupdate to capture the unthrottled 240Hz Wacom hardware rate and bypass the 60Hz display refresh latency.
- **Palm Rejection & Pen Lift Definition:** Filters out all touch events during active writing. Defines pen lift as pointerup to pointerdown gap >= 50ms.